

AmropaliNet: Detect early. Spray less. Waste less.

AI mango doctor for the Amrapali farmers of Bangladesh - protecting orchards, soil, water and bees

Problem Statement

272 words

Amrapali is Bangladesh's favourite everyday mango and the main source of income for thousands of farming families in Rajshahi, Chapainawabganj, Naogaon and the hill districts. Every season it is attacked by six common diseases: anthracnose, bacterial canker, powdery mildew, scab, sooty mould and stem-end rot.

Most farmers cannot tell these diseases apart, and an agricultural officer is rarely at hand when the first spots appear. The usual answer is blanket, calendar spraying: the whole orchard is sprayed with several fungicides and insecticides "just in case", again and again from flowering to harvest.

This turns a farming problem into an environmental one. Surplus chemicals wash into soil, ponds and groundwater, harming aquatic life and village drinking water (SDG 6). Spraying at flowering kills the bees and other pollinators that mango trees need to set fruit, and repeated doses breed resistant pathogens (SDG 15). Farm workers and their families breathe and handle these chemicals, often without protection (SDG 3).

The second problem is waste. When disease is noticed late, it has already spread, and infected fruit rots on the tree, in storage or in transport. Every discarded mango wastes the water, land, fertiliser, fuel and labour used to grow it, and rotting fruit releases greenhouse gases (SDG 12, SDG 13). Lost harvests and wasted chemical costs push small farmers towards poverty and shrink the local food supply (SDG 1, SDG 2).

ROOT CAUSE

The root cause is one missing piece of knowledge at the right moment: which disease is this, and what is the smallest treatment that works? Without it, farmers overspray, lose fruit and damage the ecosystem their orchards depend on - season after season.

Environmental harm and the SDG targets it violates

Environmental problem	SDG	Target that is violated
Chemical runoff into soil, ponds and groundwater	SDG 6	6.3 Improve water quality by reducing pollution and hazardous chemicals
Blanket, calendar spraying of agro-chemicals	SDG 12	12.4 Sound management of chemicals; less release to air, water and soil
Pollinators killed, resistant pathogens bred	SDG 15	15.5 Halt the loss of biodiversity
Farm families exposed to pesticides	SDG 3	3.9 Fewer illnesses from hazardous chemicals and pollution
Diseased fruit rots before it is eaten	SDG 12	12.3 Reduce food losses along production and supply chains
Wasted water, land, fuel and emissions	SDG 13	13 Take urgent action to combat climate change and its impacts
Lost harvests and income, less local food	SDG 2	2.4 / 1.5 Sustainable, resilient food production; resilience of the poor

Where: Rajshahi, Chapainawabganj, Naogaon and the hill districts

Who is affected: small mango farmers, farm workers, consumers, bees and orchard wildlife

AmropaliNet: the right treatment, only where needed

One photo, the right diagnosis, less chemical and less waste - a healthier environment

Solution

292 words

AmropaliNet is a free AI mango doctor, in Bangla and English, that changes every orchard decision from "spray just in case" to "treat only what is sick". Its aim: fewer chemicals in nature, less food thrown away and healthier orchards.

A farmer takes one photo of a mango with any phone. The app first checks that the photo really shows a mango, then names the disease or confirms that the fruit is healthy, and marks the affected area so the farmer can see the problem with their own eyes. Then it gives clear, practical steps.

Our environmental solution works in five ways:

- Stop needless spraying: a healthy result means no spray, breaking the habit of calendar spraying and keeping chemicals out of soil, ponds and groundwater (SDG 6, SDG 12).
- Stop wrong spraying: each disease gets its own treatment. Sooty mould, for example, grows on honeydew from sap-sucking insects, so a fungicide alone only pollutes; the app advises controlling the insects instead.
- Protect pollinators: non-chemical steps come first - pruning, orchard hygiene and neem oil, which is gentler on bees - keeping flowering orchards safe for pollinators (SDG 15).
- Cut food waste: early detection stops disease spreading from tree to tree, and hot-water treatment after harvest reduces rot in storage, saving the water, land and energy behind every mango (SDG 12, SDG 13).
- Protect people and income: less spraying means less chemical exposure for farm families (SDG 3), while lower costs and saved harvests steady their income and food supply (SDG 1, SDG 2).

It works on low-cost phones and helps extension workers reach many orchards. Every photo becomes one better decision, and season after season those decisions give soil, water and wildlife room to recover. Detect early. Spray less. Waste less.

How everything connects: from one photo to the SDGs

